**This assignment will not be graded and is only for practice.**

## Key points: 1

The span of a set $S$ (of vectors) is defined as the set of all finite linear combinations of elements (vectors) of $S$ and denoted by $Span(S)$ i.e., $Span(S) = \{\sum_{i=1}^{n} a_i v_i \in V \mid a_1, a_2, ..., a_n \in \mathbb{R} \text{ and } v_1, v_2, ..., v_n \in S\}$.
Let $V$ be a vector space. A set $S \subseteq V$ is spanning set for $V$ if $Span(S) = V$.

**Algorithm to obtain a spanning set for a non-zero vector space V:**

i) Start with the empty set $S_0 = \phi$.

ii) Add any non-zero vector $v_1 \in V$ to $S_0$ and define it as $S_1$. So $S_1 = S_0 \cup \{v_1\}$.

iii) If $Span(S_1) = V$, then we are done. Otherwise choose $v_2 \in V \setminus span(S_1)$ and add it to the set and call it $S_2$. So $S_2 = S_1 \cup \{v_2\}$.

iv) If $Span(S_2) = V$, then we are done. Otherwise, choose $v_3 \in V \setminus Span(S_2)$, and construct $S_3 = S_2 \cup \{v_3\}$.

v) Repeat the process until $Span(S_n) = V$ for some $n$.

1) Let $S = \{(1,2),(2,3),(1,-1)\} \subseteq \mathbb{R}^2$. Which of the following is (are) elements of $Span(S)$? ***1 point***

- [ ] 5(1,2) + 8(1,-1)
- [ ] (0,0)
- [ ] $2(1,2) - 3(2,3) + 5(1,-1)$
- [ ] $(2,1) - 2(3,2) - 4(-1,1)$

No, the answer is incorrect.
Score: 0

Accepted Answers:

5(1,2) + 8(1,-1)
(0,0)
$2(1,2) - 3(2,3) + 5(1,-1)$
$(2,1) - 2(3,2) - 4(-1,1)$

2) Let $S = \{(1,1,-1),(-1,1,1)\} \subset \mathbb{R}^3$. Which of the following is (are) elements of $Span(S)$? ***1 point***

- [ ] 2(1, 1, -1)+ (-1, 1, 1)
- [ ] (8, 8, -8)
- [ ] (0, 2,0)

- [ ] (1,1,-2)

No, the answer is incorrect.

Score: 0

Accepted Answers:

2(1, 1, -1)+ (-1, 1, 1)
(8, 8, -8)
(0, 2,0)

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3) Let $S$ = {(-1, 1, 1), (0, $\frac{1}{2}$ ,0), (0, 1, -2)}$\subset \mathrm{R}^3$. Which of the following option(s) is (are) true? ***1 point***

[**Hint:** Form the matrix using the vectors as the columns and calculate the determinant of the matrix.]

- [ ] $S$ is a linearly dependent set.
- [ ] $S$ is a linearly independent set.
- [ ] $S$ is a basis of the vector space $\mathrm{R}^3$.
- [ ] None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$S$ is a linearly independent set.
$S$ is a basis of the vector space $\mathrm{R}^3$.

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4) Consider a set of vectors $S = \{(1,2),(-1,2),(3,1),(-3,1)\}$ in $\mathrm{R}^2$. Choose the set of correct options. ***1 point***

- [ ] The set $\{(1,2),(-1,2),(3,1)\}$ is a basis of $\mathrm{R}^2$.
- [ ] The set $\{(-1,2),(3,1)\}$ is a basis of $\mathrm{R}^2$.
- [ ] The set $\{(3,1),(-3,1)\}$ is a basis of $\mathrm{R}^2$.
- [ ] None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The set $\{(-1,2),(3,1)\}$ is a basis of $\mathbb{R}^2$.
The set $\{(3,1),(-3,1)\}$ is a basis of $\mathbb{R}^2$.

**Key points: 2**

The following conditions are equivalent for a set to be a basis of a vector space:
- Maximal linearly independent set : A linearly independent set such that if any vector is appended to it, the new set is no longer linearly independent.
- Minimal spanning set : A spanning set such that if any vector is removed from the set, then the set will not be a spanning set.

5) Which of the following sets are maximal linearly independent sets in $\mathbb{R}^2$? *1 point*

- [ ] $\{(1,0)\}$
- [ ] $\{(1,2),(2,4)\}$
- [ ] $\{(1,2),(1,1)\}$
- [ ] $\{(1,2),(1,1),(2,1)\}$
- [ ] $\{(1,0),(0,1)\}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\{(1,2),(1,1)\}$
$\{(1,0),(0,1)\}$

6) Which of the following sets are minimal spanning sets of $\mathbb{R}^3$? *1 point*

- [ ] $\{(1,0,0),(0,1,0)\}$
- [ ] $\{(1,0,0),(1,1,0),(1,1,1)\}$
- [ ] $\{(1,0,1),(0,1,1),(1,1,0)\}$

- [ ] $\{(1,0,1),(0,1,1),(1,1,2)\}$
- [ ] $\{(1,0,0),(0,1,0),(0,0,1),(1,1,1)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(1,0,0),(1,1,0),(1,1,1)\}$
$\{(1,0,1),(0,1,1),(1,1,0)\}$

## Key Points: 3

**Basis:** A basis of a vector space $V$ is a linearly independent subset of $V$ that spans $V$.
Basis of a vector space:

- Maximal linearly independent set.
- Minimal spanning set.

How to find a basis of a subspace of $\mathbb{R}^n$?

- Suppose a subspace $V$ of $\mathbb{R}^n$ is defined by some constraints, as follows:

$$V = \{(x_1, x_2, \ldots, x_n) \mid f_1(x_1, x_2, \ldots, x_n) = 0, f_2(x_1, x_2, \ldots, x_n) = 0, \ldots, f_m(x_1, x_2, \ldots, x_n) = 0\} \subseteq \mathbb{R}^n$$

Where $f_i$'s are linear equations in general.

- **Step 1**: By solving $f_i$'s we have to try to find out the dependent and independent variables. Suppose there are $k$ independent variables and there are $k - n$ dependent variables.

- **Step 2**: Choose the first independent variable, let it be $x_1$ and assign the value 1 to that, and assign 0 to all the other independent variables.

- **Step 3**: Find out the values of all the dependent variables with respect to the assignment given in Step 2 and call that vector to be $v_1$.
- **Step 4**: Carry out this process for each independent variable, to get the vectors $v_2, \ldots, v_k$.

The set of vectors $\{v_1, v_2, \ldots, v_k\}$ is a linearly independent set and this is a maximal linearly independent set in $V$, hence forms a basis of $V$. Hence, dimension of $V$ is $k$ in this case.

7) ***1 point***

Let $S$ ={(-1, 1, 1), (0, $\frac{1}{2}$ ,0), (0, 1, -2)}$\subset \mathrm{R}^3$. Which of the following option(s) is (are) true?

**Hint:** Form the matrix using the vectors as the columns and calculate the determinant of the matrix.

- [ ] $S$ is a linearly dependent set.
- [ ] $S$ is a linearly independent set.
- [ ] $S$ is a basis of the vector space $\mathrm{R}^3$.
- [ ] None of the above.

No, the answer is incorrect.
Score: 0

Accepted Answers:

$S$ is a linearly independent set.
$S$ is a basis of the vector space $\mathrm{R}^3$.

8) Consider a set of vectors $S = \{(1,2),(-1,2),(3,1),(-3,1)\}$ in $\mathrm{R}^2$. Choose the set of correct options. ***1 point***

- [ ] The set { (1, 2) ,(-1, 2), (3, 1)} is a basis of $\mathrm{R}^2$.
- [ ] The set { (-1, 2), (3, 1) } is a basis of $\mathrm{R}^2$.
- [ ] The set { (3, 1), (-3, 1) } is a basis of $\mathrm{R}^2$.
- [ ] None of the above.

No, the answer is incorrect.
Score: 0

Accepted Answers:

The set { (-1, 2), (3, 1) } is a basis of $\mathrm{R}^2$.
The set { (3, 1), (-3, 1) } is a basis of $\mathrm{R}^2$.

9) Let $V$ be a vector space which is defined as follows: $V = \{(x, y, z) \mid 2x + 3y = 0 \text{ and } y + 5z = 0\} \subseteq \mathbb{R}^3$ with usual addition and scalar multiplication. Which of the following sets form bases of $V$? (More than one option may be correct) ***1 point***

[Hint: Two constraints are given here, $f_1(x, y, z) = 2x + 3y = 0$ and $f_2(x, y, z) = y + 5z = 0$. Observe that $z$ is the independent variable and $x$ and $y$ are the dependent variables.]

- [ ] $\{(2, 3, 0), (0, 1, 5)\}$
- [ ] $\{(3, -2, 0), (0, 5, -1)\}$
- [ ] $\{(\frac{15}{2}, -5, 1)\}$
- [ ] $\{(15, -10, 2)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(\frac{15}{2}, -5, 1)\}$
$\{(15, -10, 2)\}$

10) Let $V$ be a vector space which is defined as follows: $V = \{(x, y, z, w) \mid x + z = y + w\} \subseteq \mathbb{R}^4$ with usual addition and scalar multiplication. Which of the following set forms a basis of $V$? ***1 point***

[Hint: The constraint given here is $x + z = y + w$, which can be written as $x - y + z - w = 0$. As there are 4 variables and 1 equation, the number of independent variables is 3, where as the number of dependent variable is 1. If you check the first vector of each option, only Option 2 and Option 4 satisfying the constraint, and in that vector $x = y = 1$. Start with taking $y$ as the dependent variable, and apply the method mentioned above.]

- ○ $\{(1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 0, 1)\}$.
- ○ $\{(1, 1, 0, 0), (0, 1, -1, 0), (0, -1, 0, 1)\}$.
- ○ $\{(1, -1, 0, 0), (0, 1, 1, 0), (0, -1, 0, 1)\}$.
- ○ $\{(1, 1, 0, 0), (0, 1, 1, 0), (0, -1, 0, 1)\}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(1,1,0,0),(0,1,1,0),(0,-1,0,1)\}$.

11) Find the dimension of the vector space $V = \{(x,y,z,w) \mid x+y = z+w, x+w = y+z, \text{ and } x,y,z,w \in \mathbb{R}\}$. [Hint: Two constraints are given here $f_1(x,y,z,w) = x+y-z-w = 0$ and $f_2(x,y,z,w) = x-y-z+w = 0$. Try to find out the number of independent variables.]

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

***1 point***

**Key Points: 4**

How to find a basis of a subspace $V$ of $M_{m\times n}(\mathbb{R})$:

- **Step 1**: Consider each element of a $m \times n$ matrix as a variable. So there are $mn$ number of variables.

- **Step 2**: Using the constraints given as the definition of the subspace separate out the independent and the dependent variables. Suppose there are $k$ independent variables and $mn - k$ dependent variables.

- **Step 3**: Define a matrix $M_1$ by assigning 1 to one independent variable and 0 to all the other independent variables, together with finding out the values of all the dependent variables.

- **Step 4**: Repeat this process for all the independent variables to get the vectors $M_2, M_3, \ldots, M_k$.

The set of matrices $\{M_1, M_2, \ldots, M_k\}$ forms a basis of $V$. Hence the dimension of $V$ is $k$ in this case.

12) Let $V = \{A \mid A \in M_{3\times3}(\mathbb{R}), A \text{ is a diagonal matrix } \}$. Which of the following sets form a basis of $V$? ***1 point***

[Hint: $a_{ij} = 0$ if $i = j$. Thus there are 6 constraints and these are the dependent variables while the diagonal entries are the 3 independent elements.]

- $\left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$
- $\left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$
- $\left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$
- $\left\{ \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} \right\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$

13) Let $V = \{A \mid A \in M_{3\times3}(\mathbb{R}), A$ is a diagonal matrix and sum of the diagonal elements are 0 $\}$. What will be the dimension of $V$?

[Hint: The general form a diagonal matrix of order 3 is $\begin{bmatrix} a_1 & 0 & 0 \\ 0 & a_2 & 0 \\ 0 & 0 & a_3 \end{bmatrix}$. Hence there are three variables, $a_1, a_2, a_3$. Now there is one more constraint, which says that the sum of the diagonal elements is $0$. Hence we have $a_1 + a_2 + a_3 = 0$. So we can choose $a_3$ to be the dependent variable and $a_1$ and $a_2$ to be the independent variables, as $a_3 = -a_1 - a_2$.]

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 2

***1 point***

**Key points: 5**

Rank of a matrix:
- Maximum number of linearly independent column vectors (or row vectors) in a matrix.

NOTE: Rank of a matrix is always less than or equal to the number of columns (or rows).

How to find rank of a matrix A:
- Reduce the matrix to row echelon form.
- Find the number of non zero rows in the reduced matrix which will be the rank of the matrix A.

14) Consider the following two matrices: ***1 point***

P = $\begin{bmatrix} 1 & 0 & 3 \\ 2 & -2 & 0 \\ 0 & 1 & -1 \\ 0 & 2 & 1 \end{bmatrix}$; Q = $\begin{bmatrix} 1 & 1 & -2 & 0 \\ 1 & 0 & 2 & 0 \\ 0 & -1 & 0 & 4 \end{bmatrix}$

Which of the following statements is (are) TRUE? [ Hint: Try to reduce the matrices to row echelon form and find the number of non zero rows ]

- [ ] Rank(P) = 4
- [ ] Rank(P) = Rank(Q)
- [ ] Rank(QP) = 4
- [ ] Rank(PQ) = Rank(QP)

No, the answer is incorrect.
Score: 0

Accepted Answers:

Rank(P) = Rank(Q)

Rank(PQ) = Rank(QP)

**Your score is: 0/14**